

Zixi Yi

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Central South University | Mathematics and Applied Mathematics | Changsha, Hunan, China

Research Interests

Numerical PDEs and scientific computing; spectral methods; geometric analysis; discrete differential geometry; surface parameterization; mathematical imaging.

Education

Central South University

Sep. 2023–Present

B.S. candidate, Mathematics and Applied Mathematics

GPA: 4.0/4.0; major rank: 1/54

Selected coursework: Statistics II (99/100, class high); Analysis (98/100); Differential Equations (97/100); Computer Algebra and Dynamical Systems (97/100, class high); Fundamentals of Scientific Computing (97/100); Differential Geometry (95/100).

Graduate coursework in progress: Functional Analysis; Partial Differential Equations.

Research Experience

Central South University

Sep. 2025–Present

Research project with Prof. Yufeng Xu, Computational Mathematics

- Developing and comparing moving-mesh and rescaling schemes for semilinear reaction–diffusion equations with exponential nonlinearities and finite-time blow-up.
- Studying adaptive resolution, blow-up-time estimation, and Laplacian eigenfunction expansions as a spectral view of the evolution of spatial-frequency modes.

The Chinese University of Hong Kong (CUHK)

Jun. 2026–Present

Summer research with Prof. Gary P. T. Choi, Department of Mathematics

- Developed a conditioning-aware finite spectral framework for spherical parameterization of CT-derived genus-zero surfaces.
- Analyzed how spectral truncation and representation choices affect curvature and Laplace–Beltrami computations, motivating my extension from spectral geometry to nonlinear PDEs.

Central South University

Sep. 2024–Aug. 2025

Research project in medical imaging and machine learning

- Built and evaluated a semisupervised CT-radiomics pipeline for axillary lymph-node metastasis prediction in a multicenter cohort.
- Combined labeled and unlabeled data and assessed generalization on independent validation cohorts; published in *eClinicalMedicine* (2025).

Central South University

Sep. 2023–Aug. 2024

Research project in computational imaging

- Applied 3D CT visualization, region-of-interest construction, radiomic feature extraction, and unsupervised learning to axillary lymph-node assessment.
- Contributed to feature analysis and model validation; published in *International Journal of Surgery* (2024).

Publications

1. L. Qu, X. Mei, **Z. Yi**, Q. Zou, Q. Zhou, D. Zhang, M. Zhou, L. Pei, Q. Long, J. Meng, H. Zhang, Q. Chen, and W. Yi, “An unsupervised learning model based on CT radiomics features accurately predicts axillary lymph node metastasis in breast cancer patients: diagnostic study,” *International Journal of Surgery*, 110(9):5363–5373, 2024. doi:10.1097/JS9.0000000000001778.
2. L. Qu, J. Zhu, X. Mei, **Z. Yi**, N. Luo, S. Yuan, X. Liu, M. Liu, H. Xie, X. Hu, L. Pan, Q. Liang, Y. Li, Q. Zou, Q. Zhou, D. Zhang, M. Zhou, L. Pei, K. Qian, Q. Long, Q. Chen, X. Chen, J. K. Plichta, Q. Shang, M. Ouyang, J. Xu, and W. Yi, “Evaluating axillary lymph node metastasis risks in breast cancer patients via Semi-ALNP: a multicenter study,” *eClinicalMedicine*, 85:103311, 2025. doi:10.1016/j.eclim.2025.103311.

Manuscripts and Preprints

1. **Z. Yi** and G. P. T. Choi, *Conditioning-Aware Surface Parameterization for Finite Spectral Geometry*. Manuscript.
2. **Z. Yi**, L. Qu, and G. P. T. Choi, *An Integrated Geometric Quantification and Shape Analysis Framework for Axillary Lymph Node Metastasis in Breast Cancer Patients*. Preprint.

Honors and Awards

Central South University Academic Scholarship, **First Prize**

2024–2025

National English Competition for College Students (NECCS), Type C — **Second Prize**

2024

Central South University Academic Scholarship, **Third Prize**

2023–2024

Additional Experience

Hengrui Pharmaceuticals

Jul. 2025–Aug. 2025

Statistics Department Intern

- Applied survival analysis, Cox and logistic regression, and statistical testing to clinical data.

Standardized Tests

GRE General Test (Oct. 26, 2025): Verbal **157** (72nd percentile); Quantitative **167** (75th percentile); Analytical Writing **3.5** (40th percentile). IELTS: **7.5** overall (Listening 9.0, Reading 8.5, Writing 6.5, Speaking 6.0).

Last updated: September 2026.